

MERIDIAN FAMILY OFFICE · COPILOT v10

AI Copilot — Test Suite & Results

Verification of the deterministic engine, the deliverable layer, and the end-to-end app

108/108 tests passed · 16 July 2026

Runs fully offline in DEMO_MODE — no API key, no network, no billing.

Summary

The suite verifies one central invariant: **every number is computed deterministically; documents and instructions are advisory context that shape prose, never figures.**

Layer	What it proves	Tests
Engine	Parsing, reconciliation, suitability policy, grounded Q&A	30
Deliverable	Document ingestion, portable prompt, narrative, deck render	23
App	The Streamlit app end-to-end across every view	16
Business	Client macro views (rates, war, tax) shape prose, not the numbers	39

Statement ingestion · 9 tests

Engine Parse the three custodian formats, reconcile against each statement's own total, convert FX to USD, and keep a PII-free continuity key. Every downstream number rides here.

file: tests/test_statement_parser.py

#	Scenario verified	Result
1	Parse custodian A (USD, per-share CSV) into valued positions.	PASS
2	Semicolon CSV, European decimals, EUR/CHF → converted to a USD base.	PASS
3	JSON custodian carries illiquid alternatives + a Lombard loan (a liability).	PASS
4	One sample deliberately fails its checksum by ~\$436 — it must be flagged, not silently trusted.	PASS
5	The ~\$436 break on custodian B reconciles under a generous tolerance but fails under a tight one — proving the checksum knob actually gates the flag.	PASS
6	Continuity key is deterministic across re-parses and carries no PII.	PASS
7	Continuity check compares month-to-month keys.	PASS
8	Consolidate all three custodians into one holdings list.	PASS
9	gross – debt = net, and gross is the sum of position base values.	PASS

Suitability & policy · 12 tests

Engine The deterministic policy checks. The mandate decides what a breach MEANS; plus concentration, liquidity, unhedged-FX, hard regulatory blocks and instruction-diff.

file: tests/test_suitability.py

#	Scenario verified	Result
1	A single overweight equity book, evaluated under each mandate, yields the mandate-appropriate enforcement on the band breach.	PASS
2	A balanced book inside every band produces no actionable breach.	PASS
3	A single stock above the 10% single-security cap is caught; a diversified fund of the same weight is not.	PASS
4	Parked liquidity (money-market) is exempt from the single-position cap.	PASS
5	Below the min-liquid floor (here mostly illiquid alternatives) is flagged.	PASS
6	Economic FX exposure above the max-unhedged cap is flagged.	PASS
7	Sanctioned instrument hard blocks even execution only.	PASS
8	Complex product appropriateness blocks when not allowed.	PASS

9	Client asked for a treasury sleeve; the proposal holds a corporate-credit bond fund of the wrong duration → a deviation flag.	PASS
10	Instruction diff satisfied when matched.	PASS
11	Worst enforcement ranking.	PASS
12	The three-custodian sample book, under its advisory profile, lands on a 'flag' gate (bands breached, nothing hard-blocked).	PASS

Conversational Q&A · 9 tests

Engine Routing picks the right deterministic tool from the phrasing; answers stay grounded in computed facts; the router never crashes on odd input.

file: tests/test_portfolio_qa.py

#	Scenario verified	Result
1	'Net worth' routes to the totals tool.	PASS
2	An FX / currency question routes to the currency tool.	PASS
3	A suitability / mandate question routes to the suitability tool.	PASS
4	A 'largest position' question routes to the concentration tool.	PASS
5	A liquidity question routes to the liquidity tool.	PASS
6	A reconciliation question routes to the data-quality tool.	PASS
7	Every answer carries facts tied to the computed book.	PASS
8	The deterministic briefing fact-sheet spans every domain.	PASS
9	The router must always resolve to SOME tool — never crash on odd input.	PASS

Document ingestion · 7 tests

Deliverable The research / other uploaders read text deterministically so content can be folded into the prompt as advisory context — never a source of figures.

file: tests/test_doc_extract.py

#	Scenario verified	Result
1	Plain text and Markdown read cleanly.	PASS
2	HTML is stripped to readable text (tags removed).	PASS
3	An .eml email body is extracted.	PASS
4	CSV and JSON pass through as text.	PASS
5	An unsupported file type fails gracefully (no crash).	PASS
6	Very large documents are truncated (MAX_CHARS) and marked, never dropped.	PASS
7	The extraction result always has the same shape.	PASS

Prompt & narrative · 10 tests

Deliverable The portable prompt bundles every input; the keyless commentary REFLECTS the documents (the v10 fix) while only ever quoting the computed numbers.

file: tests/test_narrative.py

#	Scenario verified	Result
1	The prompt carries every block (FACTS, intake, holdings, docs, tactical).	PASS
2	Document text is embedded verbatim in the prompt.	PASS
3	Without documents, that content is absent from the prompt.	PASS
4	The FACTS block is valid JSON and holds the numbers.	PASS
5	v10 fix: documents change the deterministic commentary.	PASS

6	The commentary quotes short excerpts of the tactical text and docs.	PASS
7	The keyless summary must use the model's own numbers verbatim and never invent a figure that isn't already somewhere in the computed model.	PASS
8	The commentary carries the 'not investment advice' disclaimer.	PASS
9	The prompt that generates the commentary slide must NOT ask the model to build a deck / produce files — otherwise the model returns the slide-by-slide spec, which leaks onto the commentary slide.	PASS
10	The portable whole-deck prompt (a separate artifact) still asks for the full deck — that behaviour is intentional and must stay distinct from the commentary.	PASS

Proposal deck render · 6 tests

Deliverable *HTML / PPTX / PDF renderers agree, figures are copied from the model, and documents change every exported format.*

file: tests/test_generate_proposal.py

#	Scenario verified	Result
1	All three renderers (HTML/PPTX/PDF) produce valid output.	PASS
2	The deck shows the computed net-worth & gross figures.	PASS
3	The commentary slide appears only when a narrative is set.	PASS
4	v10 fix: documents change the HTML, PPTX and PDF exports.	PASS
5	With documents, the deck surfaces the client's liquidity event.	PASS
6	The deck carries the 'not independently verified' disclaimer.	PASS

End-to-end app · 16 tests

App *Drive the real Streamlit app across the views a banker walks through: load samples, review each view, change the mandate & tactical instructions — and never error.*

file: tests/test_app_scenarios.py

#	Scenario verified	Result
1	The app boots with no book loaded.	PASS
2	Each sample statement loads into the Overview view. (3 variants)	PASS
3	Overview / Holdings / Suitability / Proposal all render. (4 variants)	PASS
4	Proposal does not auto generate commentary.	PASS
5	Generate button produces commentary on click.	PASS
6	The Proposal view shows the assembled prompt and commentary.	PASS
7	Tactical instructions flow into the on-screen prompt.	PASS
8	Under discretionary, band breaches HARD-BLOCK; under advisory they only flag. The same book therefore reads differently on the Suitability view. (2 variants)	PASS
9	A single-custodian book still produces a proposal.	PASS
10	The Proposal view with no book degrades gracefully.	PASS

Business scenarios · 39 tests

Business *What happens to the output on a client macro view — a rate hike/cut, a war (or its end) prompting alternatives, a tax rise/cut: it reaches the AI, shapes the commentary, and never moves the grounded figures.*

file: tests/test_business_scenarios.py

#	Scenario verified	Result
1	A client macro view (rate/war/tax) reaches the AI verbatim. (12 variants)	PASS

2	The macro view shapes the advisory commentary. (12 variants)	PASS
3	A macro opinion shapes prose, never figures: the FACTS block (every number the deck shows) is byte-for-byte identical with and without the scenario. (12 variants)	PASS
4	'What if rates are hiked?' → the deterministic tool states the fixed-income exposure but refuses to estimate a +100bp impact it cannot source.	PASS
5	A rate-move question routes to the rate tool.	PASS
6	The scenario moves the deck's prose, not its tables.	PASS

How to run

```
python3 -m pytest                # whole suite
```

```
python3 -m pytest tests/test_suitability.py -v
```

```
python3 -m pytest -o addopts='' -v      # per-test PASS lines
```